DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

PROPOSED

EVALUATION SCHEME & SYLLABI

FOR

B. TECH. THIRD YEAR

HANDLOOM & TEXTILE TECHNOLOGY

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2024-25]

B. Tech. (HTT) Third Year, Semester- V

S.N	Subject Code	Subject Name	Type	Category	Period			Evaluation Scheme					Credit
								Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total	
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	**BHT-501**	Woven Fabric Manufacture	**T**	PSC	3	1	0	20	10	**30**	**70**	**100**	**4**
2.	**BHT-502**	Chemical Processing of Textiles – II	**T**	PCS	3	1	0	20	10	**30**	**70**	**100**	**4**
3.	**BHT-503**	Quality Evaluation of Fibers and Yarns	**T**	PCS	3	1	0	20	10	**30**	**70**	**100**	**4**
4.	**BHT051-053**	Departmental Elective– I[#]	**T**	PCS	3	0	0	20	10	**30**	**70**	**100**	**3**
5.	**BHT054-056**	Departmental Elective– II[#]	**T**	PCS	3	0	0	20	10	**30**	**70**	**100**	**3**
6.	**BHT-551**	Woven Fabric Manufacture – Lab.	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
7.	**BHT-552**	Chemical Processing of Textiles - II Lab	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
8.	**BHT-553**	Fibre and Yarn Quality Evaluation Laboratory Lab.	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
9.	**BHT-554**	Mini Project or Internship Assessment*****	**P**		-	-	-	-	-	100	-	**100**	**2**
10.		HSMC –I Audit Group/ HSMC –II Audit Group	**T**	HSMC	2	0	0	--	--	--	--	--	**0**
Total					**17**	**3**	**6**			**400**	**500**	**900**	**23**
Minor Degree/ Honors Degree MT-2/HT-2													

*****The Mini Project or Internship (4 Weeks) conducted during summer break after IV Semester and will be assessed during V Semester

Abbreviation Used:

BS: Basic Science Course
PCS: Professional Course Subject
HS: Humanities and Social Science Course

Departmental Elective-I[#]

BHT 051: Garment Manufacturing Technology
BHT 052: Home Textiles
BHT 053**:** EthnicHandloom Fabrics

Departmental Elective-II[#]

BHT 054: Production of Manufactured Fiber
BHT 055: Knitting & Non-Woven Technology
BHT 056: Textile Reinforced Composites

[#]**Students will opt one subject from the list of Department Elective.**

B. Tech. (HTT) Third Year, Semester- VI

								Evaluation Scheme					
S.N	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW) (TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	**BHT-601**	Advanced Woven Fabric Manufacture	**T**	PSC	3	1	0	20	10	**30**	**70**	**100**	**4**
2.	**BHT-602**	Fabric Structure – II	**T**	PCS	3	1	0	20	10	**30**	**70**	**100**	**4**
3.	**BHT-603**	Fabric & Garment Quality Evaluation	**T**	PCS	3	1	0	20	10	**30**	**70**	**100**	**4**
4.	**BHT061-062**	Departmental Elective–III#	**T**	PCS	3	0	0	20	10	**30**	**70**	**100**	**3**
5.		Open Elective – I	**T**	PCS	3	0	0	20	10	**30**	**70**	**100**	**3**
6.	**BHT-651**	Textile Design & Color- Lab	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
7.	**BHT-652**	Fabric Structure -II Lab	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
8.	**BHT-653**	Fabric & Garment Quality Evaluation Lab.	**P**	PCS	0	0	2	--	50	**50**	**50**	**100**	**1**
9.		HSMC –II Audit Group/ HSMC –I Audit Group	**T**	HSMC	2	0	0	--	--	--	--	--	**0**
Total					**17**	**3**	**6**			**300**	**500**	**800**	**21**
Minor Degree/ Honors Degree MT-3/HT-3													

Abbreviation Used:

BS: Basic Science Course
PCS: Professional Course Subject
HS: Humanities and Social Science Course

Departmental Elective-III#

BHT 061: Handloom Fashion & Embroidery Technology

BHT 062: Process Control in Textiles

#Students will opt one subject from the list of Department Elective.

B. Tech. V Semester
Handloom and Textile Technology
Detailed Syllabus

BHT 501	WOVEN FABRIC MANUFACTURE	3L:1T:0P	4 Credits

Objective:

1. To enable the students to understand about the warping and sizing machines.
2. To enable the students to understand about the weaving process of power loom, including different shedding mechanisms.
3. To enable the students to understand detailed concepts related to Dobby and auxiliary motion.
4. To enable the students to understand detailed concepts related to Jacquard and its recent developments and calculation related to Jacquard.
5. To enable the students to understand detailed concepts related to multiple box motion and their types and production Technology.

Unit	Topics	Lectures
I	Objectives of warping, types of warping, material flow in beam warping and sectional warping machines, types of creels used in warping machines. Objectives of sizing, types of sizing machines, principle of two cylinder and multi-cylinder sizing machine, concept of modern sizing machine. Need for drawing-in and denting-in operation, manual and automatic drawing-in and denting-in, leasing, knotting and pinning process in machines.	**12**
II	General description of plain power looms, Introduction to primary, secondary and auxiliary motion of plain power looms. Tappet shedding, over and under pick motions, Beating-up Mechanism – Eccentricity of sley, Loom timing of Primary motion, Negative and positive take up motion, five wheel and seven wheel take up motion, positive & negative let-off motions.	**12**
III	Scope & limitation of tappet shedding, Detailed study of various types of tappet shedding mechanism, Scope & limitation of dobby, negative and positive dobby, right hand & left hand dobby & their pegging methods, Brief description of Knowles dobby, cross border dobby, heald reversing motion.	**12**
IV	Types of jacquards and working principles, size and figuring capacity of jacquard, single lift single cylinder Jacquard, Double lift single cylinder, double lift double cylinder, Electronic jacquard, split harness, Different system of harness tie- up. Calculation related to Jacquard, Harness and Design- set of harness, number of harness cords to each hook, casting out and size of repeat.	**12**

V	Multiple box motion and their types, Pick-at will, pick and pick motion. Detailed study of various motions of automatic looms: pirn change, shuttle change loom, mechanical warp stop motions, electro-mechanical warp stop motion. Warp protective mechanism: loose reed & fast reed, weft detection mechanism side and center weft fork motion. Loom shed calculation-efficiency, production and production planning.	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To identify the process of warping, sizing, & drawing machines and denting process.
2. To demonstrate the weaving process of power loom, including different shedding mechanisms.
3. To summarize the working of different Dobby and auxiliary motion.
4. To illustrate working principles of Jacquard, its recent developments and calculation.
5. To describe the different types of multiple box motion and loom shed calculation.

Text Books:

1. Talukdar, M. K., Sriramulu, P. K., & Ajgaonkar, D. B. (1998). Weaving: Machines, mechanisms, management. Ahmedabad, India: Mahajan Publishers.
2. R. Mark and A.T.C Robinson, Principal of weaving (1986) the textile institute of Manchester.
3. Banerjee, P.K. (2014). Principles of Fabric Formation (1st Ed.). CRC Press.
4. Lord, P. R., & Mohamed, M. H. (1992). Weaving: Conversion of yarn to fabric.
5. Prof N.N Banerjee (2000) weaving mechanism volume I & II.
6. Thomas W.Fox, M.Sc Tech, the mechanism of weaving 2007

References:

1. Nodal center for up gradation of textile education 2008(NCUTE Production) Woven fabrics production – I & II
2. Omerod, A. (1983). Modern Preparation and Weaving Machinery. London: Butterworths.
3. Booth, J. E., & Textile Institute (Manchester, England). (1977). Textile mathematics: Vol. 3. Manchester: Textile Institute.
4. M. K. Talukdar. An Introduction to winding and Warping Testing Trade Press, Mumbai, 1982
5. Majumdar, A. B. H. I. J. I. T. (2020). Principles of Woven Fabric Manufacturing.

BHT 502	CHEMICAL PROCESSING OF TEXTILES – II	3L:1T:0P	4 Credits

Objective:

1. To enable the students to understand the basic concepts about dyes and pigments.
2. To enable the students to understand the theory of dye bath assistant, chemical auxiliaries and dyeing process parameters.
3. To enable the students to understand the detailed concept related to dyeing with Acid, Basic, Metal complex dyes on textile fibres.
4. To enable the students to understand basic concepts related to methods and style of printing.
5. To enable the students to understand about the classification and objectives of finishing.

Unit	Topics	Lectures
I	Introduction to colouring materials. Definition and classification of colouring materials: Dyes, pigments and lakes. Dye-fibre interactions: physical, chemical, solid-solid and mechanical interactions. Terminologies used in dyeing: Affinity, substantivity, exhaustion, fixation, percentage of shade, expression. Assessment of colour fastness: Colour fastness to washing, light, rubbing & perspiration.	**12**
II	Dye bath assistant, chemical auxiliaries and process parameters such as temperature, pH, time and material to liquor ratio (M.L.R) in dyeing. Principles, recipe, dyeing mechanism and method of application of Direct, Reactive, Vat, Azoic, and Sulphur dyes on cellulosic fibre. Dyeing of regenerated fibres with viscose, model, and lyocel.	**12**
III	Acid, Basic, Metal complex dyes on textile fibres (Nylon, silk, wool) with effect of process parameters and function of chemicals. Principles, recipe, dyeing mechanism and application of Disperse dyes on Polyester. Various methods of dyeing Polyester such as Carrier, High temperature, High Pressure and thermosol dyeing. Dyeing of blends: Polyester/Cotton, Polyester/Viscose and Polyester/Wool.	**12**
IV	Introduction to Printing. Difference between dyeing and printing. Ingredients in the print paste, Brief introduction to methods of printing: Block and screen printing by hand, semi-automatic and fully automatic flatbed screen printing, rotary screen printing. Styles of printing: Direct, discharge and resist style. Printing process with Direct, Reactive, and Acid dyes. After treatments in printing: Steaming and curing.	**12**

V	Objective of textile finishing, Classification of finishing processes, Mechanical and chemical finishes. Brief introduction to mechanical finishing: Calendaring and sanforizing. Brief introduction to chemical finishing processes: Crease proofing, softening and stiffening, flame retardant, water proof & water repellent finishing.	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To describe various aspects of dyes, pigments and various terms related to.
2. To use dye bath assistant, chemical auxiliaries and dyeing process parameters and mechanism involved in the process.
3. To apply the knowledge in dyeing methods for different dyes like Acid, Basic, Metal complex dyes on textile fibres.
4. To define the methods and styles of printing and machines used for printing.
5. To treat the fabric with various finishes agents.

Text Books:

1. Karakar S.R., (1999). Chemical Technology in Pretreatment of Textiles: Volume 12, Elsevier.
2. Chakraborty, J. N. (Ed.). (2015). Fundamentals and practices in colouration of textiles. CRC Press.
3. Trotman E. R., "Dyeing and Chemical Technology of Textile Fibres", B.I Publishing Pvt. Ltd. New Delhi, 1994, ISBN: 0471809101 | ISBN-13: 9780471809104.
4. V. A. Shenai, (1996). Technology of Dyeing Volume VI, Sevak Publications.
5. Choudhury A.K.R., "Modern concepts of colour and Appearance", Oxford and IBH publishing Ltd., 2000, ISBN: 1578080789 | ISBN-13: 9781578080786.
6. Choudhury, A. K. R. (2022). Principles of Textile Printing. CRC Press.
7. Prayag, R. S. (1989). Technology of Textile Printing. Noyes Data Corporation.

References:

1. Schindler, W. D., & Hauser, P. J. (2004). Chemical finishing of textiles. Elsevier.
2. Choudhury, A. K. R. (2017). Principles of Textile Finishing. Woodhead Publishing.
3. Leslie WC Miles (2003), Textile Printing, Revised Second Edition, Society of Dyers and Colourists.
4. Rivlin, J. (1992). The Dyeing of Textile Fibers: Theory and practice.
5. Kaushik C.V., Joscico A.I. Chemical processing of Textiles, NCUTE

BHT 503	QUALITY EVALUATION OF FIBERS AND YARNS	3L:1T:0P	4 Credits

Objective:

1. To make the students understand the principle and method of working of equipment's used for testing of fibres and yarns.
2. To make the students understand the concepts related to tensile testing modes like CRT, CRE, CRL.
3. To make the students understand the properties of different fibers& yarn technique.
4. To make the students understand the concepts related to yarn evenness, twist and yarn strength.
5. To make the students understand the concepts related to different aspects of yarn testing.

Unit	Topics	Lectures
I	Scope and Objectives of testing of textile materials; concept of quality; Selection of samples for quality assessment – random and biased samples – squaring technique and zoning technique for fibre selection; yarn sampling - use of random numbers - sampling for various types of yarn tests- yarn strength, linear density, elongation, twist.	12
II	Moisture relation and testing: standard atmosphere condition, moisture regain and moisture content and measurement of atmospheric condition, Fiber testing: Fibre length and length uniformity measuring techniques; Fibre strength and its relation to yarn strength. Fiber bundle strength tests. HVI, AFIS	12
III	Fibre fineness – definition- comparison of various fibres – its importance in yarn manufacture; measurement techniques. Cotton fibre maturity, estimation by microscopic method - maturity ratio and index, estimation by other methods – optical, air flow, differential dyeing; its importance in spinning. Fibre trash – influence on quality; measurement – principle and estimation of micro-dust for rotor spinning. High volume instrument for total fiber quality measurement.	12
IV	Yarn Appearance and grading, yarn mass evenness parameters – measurement –nature of irregularity: - index of irregularity; limit irregularity electronic mass evenness determination – Yarn fault classification –Yarn abrasion resistance – importance and measuring technique. Yarn hairiness – importance and assessment techniques. Twist in single and ply yarns –twist	12

	direction – twist factor – twist and yarn strength; twist measurement and breaking twist angle measurement. Analysis of data and test reports such as, Classimat.	
V	Tensile characteristics – Tensile Strength Testing modes – CRT, CRE and CRL, study of strength, single yarn strength; Lea count strength product (CSP) and Corrected Count Strength Product (CCSP), elongation, work of rupture, initial modulus, work factor and yield point – determination of yield point. Stress-strain relations of natural and manmade fibres, influence of fibre structure, humidity and temperature on tensile characteristics. Time effects Study of creep phenomena. Elastic recovery and its relation to stress and strain of fibres; mechanical conditioning of fibres and its influence on elastic recovery. Load cycling and extension cycling-their effect on elastic recovery. Introduction about torsional and flexural rigidity of fibers.	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To define the scope and objectives of testing of textile materials.
2. To explain about Moisture relation and testing for textiles.
3. To describe the testing methods of fibre fineness and maturity.
4. To explain different yarn grading system and evenness testing.
5. To describe the tensile behavior of yarns.

Text Books:

1. Booth J.E., “Principle of Textile Testing”, Butterworth Publications, London, 1989
2. Saville B.P., “Physical Testing of Textiles”, Textile Institute, Manchester, 1998
3. Kothari V. K., “Testing and Quality Management”, Progress in Textile Technology Vol.1, IAFL Publications, New Delhi, 1999
4. Arindam Basu, “Textile Testing”, The South India Textile Research Association, Coimbatore.
5. P.Angappan , R.Gopalakrishna, “Textile Testing”, S.S.M. Institute Of Textile Technology Staff abd students’ Co-op.Stores Ltd ,2002

References:

1. Slater K., "Physical Testingand Quality Control", The Textile Institute, Vol.23, No.1/2/3Manchester, 1993.
2. Ruth Clock and Grace Kunz., "Apparel Manufacture- Sewn Product Analysis", Upper Sadle River Publications, New York, 2000
3. Pradip V. Mehta., "Managing Quality in the Apparel Industry", NIFT Publication, India, 1998
4. Apurba Das," Testing of Textile and fibrous materials" CRC Pulication,2024
5. Elliot B. Grover , D.S.Hamby," Handbook of textile testing and quality control", Wiley India Pvt Ltd. (2011)

BHT-051	GARMENT MANUFACTURING TECHNOLOGY	3L:0T:0P	3 Credits

Objective:

1. To enable the students to understand the basics of garment manufacturing in apparel industry.
2. To teach students about pattern making, grading, and cutting in garment production.
3. To provide knowledge of sewing aids and equipments used in garment industry.
4. To expose the students to understand the operation of Industrial sewing machine and its parts.
5. To teach students about the use of trims and accessories in garment construction.

Unit	Topics	Lectures
I	Introduction to Garments, Apparel Industry: Structure of Industry, Various Departments in the Garment Industry and their Functions , Process Flow chart, Classification of Garments, Raw Material for Garment Manufacturing, Selection of garments (Gender, Events, Festivals, occasions, profession, sessions, community), Anthropometry, Fitting Theory (8-Head &10-Head), Body Measurement.	9
II	Pattern Making; Types of Paper Pattern, Pattern Making Tools. Principles of Pattern Drafting, Commercial Pattern, Pattern Draping, Flat Pattern Technique, Pattern Grading, Types of Grading Systems, Fabric Spreading and Cutting, Marker, Marker Parameters, Types of Markers, Marker Planning, Methods of Marker Planning, Marker Efficiency, Spreading: Types, Methods, Cutting: Preparation for Cutting, Preparation of Cut works in cutting room	9
III	Sewing Aids and equipments (Measuring tools, Drafting tools, marking tools, spreading table, cutting tools and table. Selection of Needles, threads, and fabrics –Types, properties, & classification. Seams and Stitches: Classification of Seam their use, Seam Quality, Types of Stitches: (Chain, Hand, Lock, Multi-Thread Chain, Over-Edge Chain, Covering Chain), Components of garments (Collar, Yoke, Cuff, Sleeve, Placket, Pockets, Pleat & Dart) and their types.	9
IV	Sewing Machine: Classifications, Types, Parts and Functions, Single Needle Lock Stitch Machine, Double Needle Lock Stitch Machine, Special Sewing Machines- (Over lock Machine, Flat lock machine, Buttonhole Sewing Machines, Button Sewing Machine, Feed of Arm Sewing Machine, Blind Stitch Machine, Computerized Embroidery Machine).	9

V	Apparel Supporting Materials (Linings, Interlinings, Interfacing, Shoulder Pads, Waddings and Fusing), Trims and Accessories: its objectives and uses-(Buttons, Zippers, Hooks-Eye, Hook-loop (Velcro), Elastics, Laces, Labels, Shanks and Rivets) Care labeling for apparel, pressing equipment, packing, Garment defects and remedies. Concepts of Recent Development in Seamless Garment.	**9**

Outcomes:

Upon completion of the course, the students will be able: -

1. To identify the process of Garment manufacturing and classify the garments.
2. To describe the process of pattern making & drafting, spreading, cutting and marker planning.
3. To identify the tools used in sewing and classify the seams and stitches.
4. To describe the parts & functions of various sewing machines.
5. To identify the supporting materials, trims, accessories and garment defects.

Text Books:

1. T. Karthik, P. Ganessan, & D. Gopalakrishnan, "Apparel Manufacturing Technology", CRC Press Taylor & Francis (2017).
2. Carr H., and Latham B., "The Technology of Clothing Manufacture", Blackwell Science Ltd., Oxford, 1994.

References:

1. Gerry Cooklin, "Introduction to Clothing Manufacture" Blackwell Science Ltd., 1995.
2. Harrison.P.W Garment Dyeing, The Textile Institute Publication, Textile Progress, Vol .19 No.2, 1988.
3. Jai Prakash and Gaur R.K., "Sewing Thread", NITRA, 1994.

BHT-052	HOME TEXTILES	3L:0T:0P	3 Credits

Objective:

1. To enable the students to learn about Fundamentals of home textile, materials used for home textile products and factors affecting their selection of home textile.
2. To enable the students to learn about the Kitchen textiles, draperies.
3. To enable the students to learn about the bed linen and towels.
4. To enable the students to learn about the Manufacture and properties of floor coverings.
5. To enable the students to learn different terminologies related to finishing of home textile.

Unit	Topics	Lectures
I	Home Textiles: Definition, different types of home textiles, selection of fibers, Colors, Designs, factors affecting selection of home textiles, Woven & Nonwoven, Stretch covers-Cushion covers; Table Textiles-Definitions, Different types, table mats, table cloth and hand towels, selection of material, uses. Living Room Furnishing-Sofa covers, wall hangers, cushion, cushion covers, upholsteries, bolster and bolster covers. Developments in Furnishing; Type of Furnishings, Factors affecting selection of Home Furnishings.	9
II	Kitchen Textiles and Draperies: Definition - Advances in Home decoration - Draperies-Choice of Fabrics, Types- apron-dish cloth, bread bag, pot holders, hand towels, fridge cover, fridge handle covers, mixer cover, grinder covers their use Curtains-Types of Developments in Finishing of Draperies; Tucks and Pleats; uses of Drapery Rods, Hooks, Tape Rings and Pins.	9
III	Bed Linens and Towels: Bed Linen-Definition, different types of bed linens, sheets, blankets, blanket covers, comforters, comforter covers and bed spreads; Mattress - Mattress covers, pads, pillows; Made-ups in hospitals; Textiles care labeling; Towels: Types, bath robes, bead towels, napkins; Construction of towels- weave, pile height -pattern-dyeing and finishing.	9
IV	Floor covering: hard floor coverings, resilient floor coverings, soft floor coverings, Rugs, cushions pads and care labeling; Carpet Manufacture methods & Types: Tufted, Hand tufted, Needle felt, Woven & Knotted. Carpet fibers and &yarns: Wool, wool blend, nylon, polypropylene, polyester and & acrylic.	9

V	Finishes used in Home Textiles: Introduction, Thermal draperies, Protection against unpleasant odour, Temperature-regulated beddings, Anti-microbial finish, Moisture management finish, Flame retardant finish, Mite free mattresses, Nano-technology based home textile enhancements, Sensory perception technology.	**9**

Outcomes:

Upon completion of the course, the students will be able to: -

1. To define the definition and different fibers used in home textiles.
2. To identify the different types of fabrics used for upholstery, table textiles, living room furnishing, kitchen textiles, curtains and draperies.
3. To remember the definition and application of bed linen and & towels.
4. To elaborate the manufacturing of different floor coverings according to specific needs.
5. To recognize the different care for home textiles, washing methods, and testing of home textiles.

Text Books:

1. Ramesh Babu. V. & Sundaresan S., “Home Furnishing”, Woodhead Publishing India in Textiles, 2018.
2. Dr. Subrata Das, “Performance of Home Textiles (Second Edition)”, Woodhead Publishing India Pvt. Ltd, New Delhi, 2018.
3. Alexander. N. G., “Designing Interior Environment”, Mas Court Brace Covanorich, Newyork, 1972

References:

1. Krcma. R. Manual of Non-wovens, Textile Trade Press, Manchester 1993.
2. Pradip V. Mehta., "Managing Quality in the Apparel Industry", NIFT Publication, India, 1998
3. Donserkery. K. G., “Interior Decoration in India”, D. B. Taraporeval Sons and Co. Pvt. Ltd., 1973
4. Wingate I.B. & Mohler J.F., “Textile Farbics & Their Selection”, Prentice Hall Inc., New York, 1984.
5. Irsak.C, " Nonwoven Textiles" Textile Institute", Manchester, 1999

BHT-053	ETHNIC HANDLOOM FABRICS	3L:0T:0P	3 Credits

Objective:

1. To enable the students to know the importance of textures and special features of ethnic handloom fabrics.
2. To enable the students to know the richness and importance of traditional handloom fabrics.
3. To enable the students to know the specialty of Salem silk, Angavastram, Handloom Lungies, and stoles.
4. To enable the students to know the details of different shawls, carpets, durries and furnishings made up of Kannur.
5. To enable the students to know the details of handloom articles reserved for exclusive production on handlooms.

Unit	Topics	Lectures
I	Study of construction details , loom particulars, production techniques (raw material to end product) , and specialty of Kanchipuram Saree, Chanderi Saree, Balaramapuram Saree , Venkatagiri Saree, Banaras Brocade and Saree, Paithani Saree, Baluchari Saree, maheshwari saree, Jamdani, AwadhJamdani, Muga silk saree of Assam.	**9**
II	Study of construction details, loom particulars, production techniques (raw materials to end product) and Specialty of Patola Saree of Gujarat, Ikats of Andhra Pradesh and Orissa, Kalam Kari,Tie and Dye fabrics of Rajasthan-Bhandhej, Mothda&Laheriya.	**9**
III	Construction details, loom particulars, production techniques(Raw Marterial to end products) and specialty of Salem silk and cotton Dhotis, silk and Cotton Angavastram, Handloom Lungies, Study of stoles- Kinnori stole, Pashmina stole, Tangalia stole, Bagru block print stoles.	**9**
IV	Study of shawl- Himroo Shawl, Kullu Shawl , Naga Shawl , Wool Jamawar Shawls, Cashmere Check Shawl, Pashmina Shawls, Carpets – Hand Woven Pile Carpets, Plain & Figured Durries , PanipetKhes and Tapestry – Furnishing & made –ups of Kannur.	**9**

V	Study of handloom articles reserved for exclusive production on handlooms Saree, Dhothi, Towel and Gamcha, Angavastarm,Lungi, Khes, Bed sheets, Bedcover, Counterpane Furnishing (including Tapestry, Upholstery) Jamakkalam, Durry or Durret, Dress material, Barrack Blankets and Kambalor Kamblies, Shawl, Loi, Muffler, Pankhi, Woollen Tweet, Chaddar , Mekhala or Phanek.	**9**

Outcomes:

Upon completion of the course, the students will be able: -

1. To identify the different textures and special features of ethnic handloom fabrics.
2. To remember the richness and importance of traditional handloom fabrics.
3. To explain the specialty of Salem silk, Angavastram, Handloom Lungies, and stoles.
4. To illustrate different shawls, carpets, durries and furnishings made up of Kannur.
5. To interpret the different handloom articles reserved for exclusive production on handlooms.

Text Books:

1. S. Rangarajan, "HEPC product manual", Handloom Export Promotion council, Chennai,1994.
2. R.T.A Kapur Chishti&Ambasanyal, "Saris of India", Wiley Eastern Ltd, New Delhi, 1989.
3. Dr. Moti Chandra, "Ikat sarees of Orissa and Andhra pradesh", Calico museum of Textiles,Ahmedabad, 1974.
4. Amin Hirenbhai Navinbhai, Parmar Dharmendra Kumar, "Handloom Harmony: Blending tradition with Modern Expression", Archers & Elevators Publishing House (2023).
5. Amin Hirenbhai Navinbhai, Parmar Dharmendra Kumar, "Empowering Weavers: Initives for Sustanable Handloom Growth ", Archers & Elevators Publishing House (2023).

References:

1. "India Handlooms – Tantavi Tradition & Trend catalogue" NCTD, O/o. Development Commissioner for Handlooms, New Delhi, 2015.
2. B. K. Behera, Traditional Textile Designs – Gold line Publisher - New Delhi.

BHT-054	PRODUCTION OF MANUFACTURED FIBRES	3L:0T:0P	3 Credits

Objectives:

1. To enable the students to learn about the fiber forming process, fiber morphology and its application areas.
2. To expose the students to principle phenomena of fluid flow during fiber spinning.
3. To expose the students to advances in the melt spinning process.
4. To brief students to advances in the solution spinning process.
5. To enable the students to learn about the advanced post spinning operations, and high performance fibers and its applications.

Unit	Topics	Lectures
I	Evolution of Manufactured fiber, Fiber forming process, Fiber products and properties, Molecular orientation and crystallinity in fibers, Polymer as Fiber, Plastics and Rubber, Fiber Morphology, Thermal Transitions. Application area.	**9**
II	Phenomena of fluid flow during fiber spinning: Shear Flow- Viscose flow and Newtonian fluids, capillary flow. Non-Newtonian Fluids - Time-independent fluids & Time-dependent fluids, Factor affecting shear viscosity, Spinnability of fluids, Flow Instabilities – dieswell and melt fracture, draw resonance.	**9**
III	Melt Spinning: Spinning Line - Feature, extruder, spinning manifold, spin pack & spinneret, cooling system. Melt spinning variables and Conditions for Continuous Spinning, Special Features of High Speed Spinning – LOY, POY, FOY, and HOY. Structure development in Polyester and Nylon during spinning.	**9**
IV	Solution Spinning: Process variable in solution spinning. Dry-Spinning – Preparation of dope, Spinning process – spinning cell, fiber cross-section formation, spin-stretch during dry-spinning, finish application and winding. Wet-Spinning – Solution Preparation and Transport, Coagulation-coagulation process, Spin-stretch during coagulation, structure and morphology of fiber in solution-spinning, Salient features of solution spinning.	**9**
V	Post-Spinning Operations: Drawing - drawing systems, influence of drawing on structure and properties of fibres, Heat setting- Types of heat setting, influence of heat setting on fiber behavior, Spin finish - Importance of Spin finish composition & application, Texturing- classification & different techniques of texturizing. Introduction and use of high performance fibers- Aramids, Carbon, Glass.	**9**

Outcomes:

Upon completion of this course, the students shall be able:-

1. To explain the fiber forming process, fiber morphology and its application areas.
2. To describe principal phenomena of fluid flow during fiber spinning.
3. To discuss the working of the melt spinning machines.
4. To compare and explain the different types of solution spinning process.
5. To describe the advanced post spinning operations, and application of high performance fibers.

Text Books:

1. Kothari V. K., "Textile Fibres: Development and Innovations", Vol. 2, Progress in Textiles,IAFL Publications, New Delhi, 2000
2. Characterization of Polymers and Fibres: M K Singh, A Singh Elsevier Publication Cambridge USA, 2021, eBook ISBN: 9780128242391
3. Nakasjima (English edition, edited by Kajiwara K. and McIntyre J. E.), "Advanced FibreSpinning Technology", Wood head Publication Ltd., England, 1994.

References:

1. Gupta V. B. and Kothari V. K. (Editors), "Manufactured Fibre Technology", Kluwer Academic Publishers, 1997.
2. Cook J. G., "Handbook of Textile Fibres: Vol. 2: Man Made Fibres", The Textile Inst., 5th Ed. 1984.
3. Srinivasa Murthy H. V., "Introduction to Textile Fibres", Textile Association, India, 1987.
4. Uddin, F. , (Ed.). (2019). Textile Manufacturing Processes.
5. Gupta, V. B., & Kothari, V. K. (2012). Manufactured Fibre Technology. Dordrecht: Springer Netherlands.

BHT-055	KNITTING & NON-WOVEN TECHNOLOGY	3L:0T:0P	3 Credits

Objectives:

1. To enable the students to learn about different knitted fabric structure and their properties.
2. To expose the students about the elements and mechanism of different types of knitting machines.
3. To expose the students about the classification of non-woven fabric.
4. To brief students about the different manufacturing methods of non-woven bonding.
5. To enable the students to learn about production method, application of non-woven.

Unit	Topics	Lectures
I	Introduction to Knitting; Difference between woven, knitted and non-woven products and processes. Classification of knitting machines. Elements of weft knitting: needles, sinker and cam. Terms and Definitions used in knitting**:** Wales, Courses, Face loop, Back loop, Stitch density, Pitch, Gauge, knit, Float & Tuck Stitches and their effects on fabrics. Some stitch notation.	**9**
II	Weft Knitting: types of knitting needle, Circular weft knitting machine and its mechanism. Knitting action of Latch needle and bearded Needle. Classification of weft knit structures and Properties of Plain, Rib, Interlock and Purl structures, and its end uses.	**9**
III	Introduction and knitting element of warp knitting, concept of loop formation, swinging and shogging motion, warp knitting machine: tricot and Rachel warp knitting machine and their properties and applications.	**9**
IV	Introduction to non-woven technology, classification of non-woven technology, Types of fibers used in non-woven fabric, Methods of web preparation, Orientation of fibres in the web: Cross laid, parallel laid, Random laid.	**9**
V	Non-woven fabric properties influencing factors. effects of process a variables on properties of non-woven fabric. Methods of bonding of web: Needle Punching, Chemical Bonding, Thermal bonding, Spun laid technology & application of non-woven fabrics.	**9**

Outcomes:

Upon completion of this course, the students shall be able:-

1. To define the different types of knitted fabric and their properties.
2. To explain the weft knitting mechanisms and applications.
3. To explain the weft knitting mechanisms and applications.
4. To describe the manufacturing methods of non-woven bonding.
5. To illustrate the production process and application of non-woven.

Text Books:

1. B Ajgaonkar, Knitting technology, Universal Publication Corporation, Mumbai, 1998.
2. N. Anbumani, Knitting Fundamentals, Machines, Structures and Developments‖, New Age International, 2006.
3. J. Spencer, Knitting Technology, Textile Institute Manchester,2005.
4. J.Russell, "Hand Book of Nonwovens", Wood head publications Ltd., ISBN-13: 978-1- 85573-603-0,2007.
5. Wilhelm A, Hilmar F and Walter K, "Nonwoven Fabrics: Raw Materials, Manufacture, Applications, Characteristics, Testing Processes", Wileyverlag Gmbh & Co. Kgaa, Weinheim, 2003.

References:

1. J.Russell, "Hand Book of Nonwovens", Wood head publications Ltd., ISBN-13: 978-1- 85573-603-0, 2007.
2. Wilhelm A, Hilmar F and Walter K, "Nonwoven Fabrics: Raw Materials, Manufacture, Applications, Characteristics, Testing Processes", Wileyverlag Gmbh & Co. Kgaa, Weinheim.
3. R.A chapman "Application of nonwovens in technical textiles", Wooded publication limited(2010)

BHT-056	TEXTILE REINFORCED COMPOSITES	3L:0T:0P	3 Credits

Objectives:

1. To enable the students to learn about different textile-reinforced composites.
2. To break down complex composite structures into their constituent components.
3. To enable the students to learn about Manufacturing Methods and Processes of Composites.
4. To enable the students to learn about Characterization of Composites analysis.
5. To expose about the suitability of different textile-reinforced composites for specific applications based on performance requirements.

Unit	Topics	Lectures
I	Introduction to textile reinforcement composite, Definition, classification, and materials used in composites, textile composites, and textile structural composites, Single Polymer Composite, Materials for composites, High performance Fibers:-glass, carbon, aramid, boron, ceramic and natural fibers-(Jute &flax), Reinforcement structures, Classification of Textile Reinforced Structures based on axis and dimension; non-axial, mono-axial, biaxial, tri-axial, and multi-axial structures, parallel, Chopped strand, and Milled fibers	**9**
II	Matrix classification, properties and selection of matrix, Polymer matrix, Thermoset matrix – epoxy, polyester, vinyl ester. Thermoplastic matrix– polyether ether ketone, poly-phenylene sulfide, poly-sulfone. Metal matrix and Ceramic matrix,Carbon-Carbon Composites, Recycling Fiber Reinforced.	**9**
III	Manufacturing Methods and Processes of Composites - Hand and machine lay-up, spray-up molding, vacuum-bag and pressure-bag moulding, autoclave, compression moulding, liquid resin molding, resin transfer moulding, Poltrusion, filament winding, injection moulding, thermoplastic processing, automated tape lying.	**9**
IV	Characterization of Composites, Mechanical Analysis, Tensile, bending, compression, impact, fatigue behavior, Morphological characterization of composites reinforced by various textile structures.	**9**

V	Applications of Textile structural composites Automotive, marine, aerospace, sports, protection systems, wind energy, machine components, construction engineering	**9**

Outcomes:

Upon completion of this course, the students shall be able:-

1. To explain the materials and classification of textile-reinforced composites.
2. To describe the complex composite structures into their constituent components.
3. To illustrate the manufacturing methods and processes of composites.
4. To identify the characterization of Composites analysis.
5. To select the application of textile-reinforced composites for specific applications based on performance requirements.

Text Books:

1. Tsu Wei Chou and Frank K. Ko , Textile structural composites by, ELSEVIER, 1989
2. Jinlian Hu, 3D Fibrous assemblies, Woodhead Publication, 2009
3. J G Morley, Harcourt Brace Jovanovinch, High-performance fiber composites, 1987
4. S Advani, Textile structural composites, Conference proceedings, 2007
5. A Long, Design and Manufacturing of Textile structural composites, Woodhead Publication

References:

1. P.K.Mallick, Fibre-reinforced composites by CRC press.
2. A R Horrocks and S C Anand, Handbook Of Technical Textiles, Woodhead Publication

BHT-551	WOVEN FABRIC MANUFACTURE –LAB.	0L:0T:2P	1 Credits

S. No.	List of Experiments
1	To perform dismantling, erecting tappet shedding mechanism.
2	To perform dismantling, erecting and time setting over-picking mechanism.
3	To perform dismantling, erecting and time setting under picking mechanism.
4	To perform dismantling, erecting and time setting beat-up mechanism.
5	To determine the pick space through 5 and 7 wheel take-up mechanisms.
6	To prepare the pattern card for dobby shedding mechanism and way in which adjust the depth of shed.
7	To study about layout of jacquard loom and familiarization of mounting jacquard on a loom.
8	To study about different systems of harnessing.
9	To practice the harnessing building for straight tie, sectional tie, pointed tie and body border tie.
10	To study about warp protector mechanism (loose reed, fast reed).
11	To study the warp and weft stop motion.
12	To study about drop box motion of power loom.

BHT-552	CHEMICAL PROCESSING OF TEXTILES - II LAB.	0L:0T:2P	1 CREDITS

S. No.	List of Experiments
1	Dyeing of cotton yarn/fabrics with Direct dyes
2	Dyeing of cotton yarn/fabrics with Reactive dyes.
3	Dyeing of cotton yarn/fabrics with Vat dyes
4	Dyeing of cotton yarn/fabrics with Azoic dyes
5	Dyeing of silk & wool yarn/fabrics with acid dyes
6	Dyeing of Wool & silk yarn/fabrics with basic dyes
7	Dyeing of silk yarn/fabrics with Metal complex dyes
8	Direct style of Printing of Silk fabrics with Acid dyes using Blocks/Screens
9	Discharge style of printing of cotton fabrics with Reactive dyes
10	Resist style (Batik Printing) of cotton fabrics with Reactive cold brand dyes
11	Crease resistance finishing on Cotton fabrics.
12	Softening and stiffening finish on cotton fabrics

BHT-553	FIBRE AND YARN QUALITY EVALUATION- LAB.	0L:0T:2P	1 Credits

S. No.	List of Experiments
1	To Measure fibre fineness by air flow method.
2	To find out fibre length by Baer Sorter & analytical digital fibrograph.
3	To find out Fiber maturity test by Caustic Soda Swelling method.
4	To studies fibre trash by Shirley trash analyzer.
5	To find out fibre bundle strength by using Pressley fibre bundle strength tester.
6	To find out linear density of sliver, Roving and yarn.
7	To find out the yarn strength by using Single yarn strength tester.
8	Fine out yarn lea strength of yarn by using Lea strength tester.
9	To find the Single and folded yarn twist by using yarn twist tester.
10	Measurement of yarn Unevenness by using USTER evenness tester.
11	To find out Yarn appearance by using Yarn Appearance Board Winder

B. Tech. VI Semester
Handloom and Textile Technology
Detailed Syllabus

BHT-601	ADVANCED WOVEN FABRIC MANUFACTURE	3L:1T:0P	4 Credits

Objective:

1. To enable the students to understand about the production techniques Projectile Weaving Machine.
2. To enable the students to know about the production techniques Rapier Weaving Machine
3. To enable the students to know about the production techniques Air-jet and Water-jet Weaving Machine.
4. To enable the students to know about the production techniques terry weaving and multiphase weaving.
5. To enable the students to know about the calculation related to maximum setting of cloth.

Unit	Topics	Lectures
I	Projectile Weaving Machine: Basic Principle of projectile loom, sequence of weft insertion, beat-up, torsion bar picking, loom timing, let-off and take-up motion, selvedge-formation, Energy equation of torsion bar, velocity and acceleration of the projectile.	**12**
II	Rapier Weaving Machine: Method of weft insertion system- Gabler, dewas types of rapier; rigid, flexible, telescope, weft insertion sequence in rapier weaving process. Different methods to drive the rapier head. Single phase double acting rapier. Velocity of the rapier. Loom timing.	**12**
III	Air jet weaving machine: Problem in air jet weft insertion, Sequence of weft insertion in air jet loom. Design of confuser guide, design of profile reed and relay jet. Loom timing. Water-jet Weaving machine: Weft insertion mechanism, quality of warp required for water jet, quality of water, problem of water-jet loom.	**12**
IV	Introduction to terry weaving, Classification of terry fabrics, different mechanism for terry Manufacturing, Properties and performance of terry fabric. Multi-phase Weaving Machine: Basic concept of multiphase weaving: circular weaving machine.	**12**
V	Calculation related to maximum setting of cloth. Determination of Ends per inch and picks per Inch while changing count or weave and its combination to	

	maintain the same level of Compactness. Diameter of yarns Ashenhurst's & Pierce's formula for estimation of diameter of yarns. Cloth cover- Derivation and calculation. Fractional Cover, Percentage cover and Cover factor – Calculation for light medium and heavy fabrics.	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To explain the production techniques Projectile Weaving Machine.
2. To illustrate the production techniques Rapier Weaving Machine.
3. To explain the production techniques Air-jet and Water-jet Weaving Machine.
4. To describe the production techniques terry weaving and multiphase weaving.
5. To apply mathematical calculation & relation related to maximum setting of cloth.

Text Books:

1. Banerjee, P.K. (2014). Principles of Fabric Formation (1st Ed.). CRC Press. https://doi.org/10.1201/b17804.
2. Marks, R., & Robinson, A. T. C. (2022). Principles of weaving. Manchester [England: Textile Institute.
3. Lord, P. R., & Mohamed, M. H. (1992). Weaving: Conversion of yarn to fabric.
4. Talukdar, M. K., Sriramulu, P. K., & Ajgaonkar, D. B. (1998). Weaving: Machines, mechanisms, management. Ahmedabad, India: Mahajan Publishers.
5. O.Talavarsek/V.svaty shuttleless weaving machine (1981) volume 3.

References:

1. Omerod, A. (1983). Modern Preparation and Weaving Machinery. London: Butterworths.
2. Booth, J. E., & Textile Institute (Manchester, England). (1977). Textile mathematics: Vol. 3. Manchester: Textile Institute.
3. M. K. Talukdar. An Introduction to winding and Warping Testing Trade Press, Mumbai, 1982
4. Majumdar, A. B. H. I. J. I. T. (2020). Principles of Woven Fabric Manufacturing.

BHT-602	FABRIC STRUCTURE – II	3L:1T:0P	4 Credits

Objective:

1. To enable the students to understand the construction of double cloth.
2. To enable the students to understand the Tubular cloth, Treble and backed cloth.
3. To enable the students to understand the Pile fabrics, Velvet, two beams and 'W' binding.
4. To enable the students to understand the Velveteen, Corduroys, Distorted thread structures, Leno and gauze fabrics, Bottom douping and Top douping, Easer and shaker device.
5. To enable the students to understand the welts and pique, Bedford cord fabric and jacquard design.

Unit	Topics	Lectures
I	Introduction and classification of double cloth- Step by step construction of self-stitched double cloth, reversible and non-reversible using twill and sateen weave, Centre stitched double cloth using twill and sateen weave. Thread interchanging double cloth- warp thread interchanging double cloth and weft thread interchanging double cloth using plain and twill weave. Wadded double cloth–warp wadding and weft wadding interlacement diagram and its design, draft, peg plan, tie-up plan of above weaves.	**12**
II	Introduction of plain Tubular cloth, Treble width plain cloth– interlacement diagram and it's graphical representation; Treble cloth using twill, satin, and sateen weave; design, draft, peg plan, tie-up plan and thread interlacing diagram. Backed cloths – Salient features of backed cloths, Warp and weft backed cloths - Reversible and non-reversible warp backed cloth using twill and sateen; design, draft, peg plan, tie-up plan and thread interlacing diagram.	**12**
III	Pile fabrics – Salient features, classification of pile fabrics, Terry piles – salient features, classification of terry pile structures – 3 pick, 4 pick, 5 pick and 6 pick terry terry pile on one side of fabric and on both sides, stripe and check effects - design, draft, peg plan, tie-up and thread interlacing diagram of above weaves. Velvet - Warp pile fabrics produced with the aid of wires-loop piles and cut piles, classification of wire pile fabrics- all over and fast wire pile structure- design, draft, peg plan, tie-up and thread interlacing diagram of above weaves.	**12**
IV	Velveteen- salient features, manufacturing technique and processes involved; All over velveteen – Plain back and Twill back velveteen, Corduroys; design, draft, denting, peg/tie-up plan and thread interlacing diagram of above	

	weaves. Gauze and leno fabrics – Salient features, sheds form in leno weaving, Bottom douping and Top douping, Easer and shaker device; thread interchanging diagram of leno structures, Combination of Leno- plain and twill weaves, Russian Cord, Net leno -drafting and lifting plans.	**12**
V	Welt and pique structure- salient features and manufacturing technique, ordinary welts, wadded welts- loose back welt and fast back welt-design, draft, peg plan and thread interlacing diagram. Bedford cord weave- silent features, plain faced Bedford cord (pair and alternate pick principle), twill faced Bedford cord, wadded Bedford cord and crepon Bedford cord - design, draft, peg plan and thread interlacing diagram. Introduction of construction and development of jacquard designs, design development and punching process by using straight tie and straight draft, Salient feature of Damask and Tapestry fabric	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To explain the construction of different types of double cloth.
2. To elaborate the Tubular cloth, Treble and backed cloth.
3. To describe the Pile fabrics, Velvet, two beams and 'W' binding.
4. To explain the Velveteen, Corduroys, Distorted thread structures, Leno and gauze fabrics, Bottom douping and Top douping, Easer and shaker device.
5. To explain the welts and pique, Bedford cord fabric and jacquard design.

Text Books:

1. Grosicki Z. J. "Watson's Textile Design and Colour: Vo.1, Woodhead Publications, Cambride England, 2004.
2. Grosicki Z. J., "Watson's Advanced Textile Design and Colour", Vol.II, Butterworths, London, 1989.
3. Wilson J., "Handbook of Textile Design", Textile Instiute, Manchester, 2001.
4. Home C.E., "Geometric Symmetry in Patterns and Tilings" Textile Institute, Manchester, 2000.
5. Seyam A. M. "Structural Design of Woven Fabrics, Theory and Practice" Textile Institute, Manchester, 2002.

References:

1. Geormer D, "Woven Structure and Design, part 1: Single Cloth Construction: WIRA, U.K. 1986
2. Geomer D, "Woven Structure and Design, Part 2: Compound Structures" WIRA, U.K., 1989.
3. Gokarneshan, N. (2004). Fabric structure and design. New Delhi, India: New Age International (P) Ltd., Publishers.

4. Foster, J. S., &Greeno, R. (2013). Structure and fabric: Part 2.

5. Gokarneshan, N. (2009). Fabric structure and design. New Delhi, India: New Age International (P) Ltd., Publishers.

6. H.Nisbet, FTI Grammar of textile design , textile technologist & consultant.

BHT-603	FABRIC & GARMENT QUALITY EVALUATION	3L:1T:0P	4 Credits

Objective:

1. To enable the students to learn about the construction of fabrics and evaluation of fabric properties and their importance.
2. To enable the students to learn about the Tensile strength measurement, Tear strength, Bursting strength and its measurement.
3. To enable the students to learn about the Fabric stiffness, Drape ability, Crease recovery, Wrinkle recovery, Fabric abrasion resistance, Fabric pilling resistance.
4. To enable the students to learn about hysteresis; compression and decompression; surface roughness and friction measurements; Textile product leveling; Uster standards, AITCC, JIS and ASTM.
5. To enable the students to learn about Fabric inspection, classification of fabric defects, MIL standards and final inspection, Colour matching, Analysis of KES and FAST data.

Unit	Topics	Lectures
I	Basic fabric particulars – Measurement of ends and picks per inch, count of warp and weft, determination of the type of weave, measurement of length, width, thickness and Areadensity (GSM); warp and weft crimp measurements for spun and filament yarn fabrics, the cover factor calculations; Fabric sampling techniques.	12
II	Tensile strength measurement – raveled strip test and grab test – mechanical and electronic measuring systems. Tear strength – importance – measuring systems. Bursting strength and its measurement. Ballistic impact strength. Universal tensile tester - principle and operation.	12
III	Fabric stiffness – principle of measurement of flexural rigidity; Drapability – measurement of drape coefficient; Crease recovery measurement techniques. Wrinkle recovery assessment using standard grades; Principle and functioning of air permeability testers, water repellency, contact angle and fabric shrinkage testing; Fabric abrasion resistance – measuring technique; Fabric pilling resistance – methods of determination.	12
IV	Fabric bending hysteresis testing; Shear hysteresis measurements; Fabric compression and decompression behavior; Fabric surface roughness and friction measurements; Fabric tensile hysteresis measurements; Fabric	12

	flame resistance testing methods; Moisture and thermal characteristics. Textile product leveling. International quality parameters and standards like Uster standards, AITCC, JIS and ASTM.	
V	Fabric inspection – Manual, semi-automatic and Automatic Inspection systems, classification of fabric defects, independent product quality certification, acceptable quality level, MIL standards and final inspection. Quality assessment of garments - cutting, sewing, pressing, finishing and packaging defects. Assessment of comfort, Colour matching, Analysis of KES and FAST data.	**12**

Outcomes:

Upon completion of the course, the students will be able: -

1. To evaluate the constructional parameters of fabrics.
2. To analyze the tensile strength of fabrics.
3. To evaluate low stress mechanical properties of fabric.
4. To elaborate the quality parameters and standards like Uster standards, AITCC, JIS and ASTM.
5. To explain the quality assessment of fabrics and garment.

Text Books:

1. Booth J.E., "Principle of Textile Testing", Butterworth Publications, London, 1989
2. Saville B.P., "Physical Testing of Textiles", Textile Institute, Manchester, 1998
3. Kothari V. K., "Testing and Quality Management", Progress in Textile Technology Vol.1, IAFL Publications, New Delhi, 1999

References:

1. Ruth E clock and Grace Kunz., "Apparel Manufacture – Sewn Product Analysis", Upper Sadle River Publications, New York, 2000
2. Bubonia, J. E. (2014). Apparel quality lab manual.

BHT-061	HANDLOOM FASHION & EMBROIDERY TECHNOLOGY	3L:0T:0P	3 Credits

Objective:

1. To express the knowledge & use of appropriate basic design and Fashioning Tradition.
2. To understand about the difference between High fashion, Mass fashion & custom-made clothing.
3. To understand about the colour theory and concept of embroidery.
4. To learn & identify about the embroidery tools and regional embroidery.
5. To learn about the tools & equipments used in Surface Ornamentations and Special Surface Ornamentations.

Unit	Topics	Lectures
I	Concept of Handloom Act, 1985, Handloom Mark, Geographical Indication (GI) of Handloom Products, Handloom brand. Fashioning Tradition, Handloom Fashion: Fashion, Definition, Fashion Terminology, Types, Fashion Cycle (Stages of Fashion). Basic Design: introduction, definition, types of design: structural & decorative, source of inspiration of design, elements of design, principle of design. Elements of handloom arts & Fashion.	**9**
II	Elements of Fashion illustration: Introduction to Fashion illustration – Eight and twelve head theory, History, importance, artists and illustrators of national and international repute. Elements of Design (point, line, form, shape, space, size, texture and colour), Principles of Design (harmony, proportion, balance, rhythm and emphasis).	**9**
III	Colour Theory (CIE lab colour system, Pantone Colours, colour wheel, colour value scale, grey scale, colour schemes, colour psychology, colour and emotions, Indian approach to colour). Definition of Embroidery, Embroidery kits, tools, & Equipments for embroidery.	**9**
IV	Basics Stitches of hand embroidery (running stitch, backstitch, split stitch, stem stitch, satin stitch, herringbone and French knots), m/c embroidery, computerized embroidery. Regional embroidery: Study of motifs, fabrics, colour threads, stitches used for design in Indian state of Traditional embroidery (Chikankari, phulkari, Kantha, Chandua, banjaara.) Concise Introduction of Aari Work & Zardozi Work.	**9**
V	Some Special Surface Ornamentation: Gold and Silver thread work – Materials and Methods used, Mirror Work, Raised Work Appliqué, Quilting,	

	Patch Work, Tapestry weaving: Principles, appliances and materials, bobbins & needles, Comb, embroidery frames treated as a loom.	**9**

Outcomes:

Upon completion of the course, the students will be able: -

1. To define the appropriate handloom fashion and design terminology as per standards.
2. To compare difference between High fashion, Mass fashion & custom-made clothing.
3. To explain the colour theory and concept of embroidery.
4. To describe the embroidery tools and regional embroidery.
5. To identify the tools & equipments used in surface ornamentations and Special Surface Ornamentations.

Text Books:

1. Amin Hirenbhai Navinbhai, Parmar Dharmendra Kumar, "Handloom Harmony: Blending tradition with Modern Expression", Archers & Elevators Publishing House (2023).
2. Amin Hirenbhai Navinbhai, Parmar Dharmendra Kumar, "Empowering Weavers: Initives for Sustanable Handloom Growth", Archers & Elevators Publishing House (2023).
3. Colour Harmony, Bride N. Whelan, Rockport Publishers..
4. Embroidery Designs by Nirmala C Mistry ,Navneet Publisher

References:

1. Soamn, Jullian, "Professional Fashion illustration" B.T. Batslord, London 1995 .
2. Drake, Nicholas, "Fashion illustration today" Thamesis Hudson. London Publication
3. Encyclopedia of embroidery by Reader digest
4. Traditional Indian textiles-John Gillow

BHT 062	PROCESS CONTROL IN TEXTILES	3L:0T:0P	3 Credits

Objectives:

1. To enable the student to know about process control and quality assessment of Pre-Spinning
2. To enable the student to know about process control yarn productions.
3. To enable the student to know about process of winding and warping in weaving Process.
4. To enable the student to know about process control and quality assessment of Sizing.
5. To enable the student to know about process control in Textile Chemical Processing.

Unit	Topics	Lectures
I	Importance of Process control in textiles. Control of mixing quality and cost, determination of trash content and cleaning efficiency of blow room and card , control of neps generation and fibre rupture in blow room and card, yarn quality characteristics affected by blow room and card, neps removal efficiency (NRE) in carding, factor affecting NRE in carding. control of process parameter in blow room and carding.	9
II	Influence of roller setting, construction of top and bottom roller and draft distribution on yarn characteristics, optimum level of comber waste, improvement in mean length in combing, maximizing machine efficiency in ring frame, controlling the end breakage rate in ring frame, control of yarn realization. Control of count variability, unevenness, imperfection and yarn faults.	9
III	Process Control in Winding: characteristics of good splice-appearance and strength rating, yarn faults classification by Classimate, yarn clearer setting adjustment and removal of yarn faults, package defects, Relative humidity and temperature. Process Control in warping: Control of end breaks, tension levels, quality. Process Control in Pirn Winding: Minimising End breaks, stoppages due to mechanical failure.	9

IV	Process Control in Sizing: Choice of size recipe and preparation, control of size pick-up, control of yarn stretch in different zones, Control of moisture in sized yarn. Control of Productivity in loom shed, control of loom efficiency and stoppages.	**9**
V	Process control in textile pre-treatment - Desizing, Scouring, Bleaching, Souring, Mercerizing. Process control in dyeing and printing of textiles- Basic requirements of water quality for dyeing, Influence of Liquor ratio, auxiliaries, temperature, pH in reactive dyeing, Process control in printing paste formulation. Process control in finishing of textiles- Stenter, sanforising, Calendaring process, and Chemical finishes- Softener, Resin, Fire retardants, and water repellent and finish.	**9**

Outcomes:

Upon completion of this course, the students shall be able:-

1. To explain the process control in Pre-Spinning.
2. To describe the process control in yarn productions.
3. To explain the process control in winding and warping in weaving Process.
4. To illustrate about the process control in Sizing process in textile manufacturing.
5. To interpret about process control in Textile Chemical Processing.

Text Books:

1. Abhijit Majumdar, Apurba Das, R. Alagirusamy and V. K. Kothari, "Process control in textile manufacturing" Woodhead Publishing Limited, (2013)
2. M C Paliwaland P D Kimothi, "Process control in weaving" ATIRA Publication

References:

1. G.Thilagavathi and T. Karthik "Process control and yarn quality in spinning" (2016) by Woodhead Publishing India Pvt. Ltd.
2. Garde A.R. and Subramaniam T.A., "Process Control in Spinning", ATIRA Publicaitons, Ahmedabad, 1989
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6. Volz.H.Z, "Industrial Colour Testing – Fundamentals and Techniques", VCH, 1994

BHT-651	TEXTILE DESIGN & COLOR- LAB	0L:0T:2P	1 Credits

S. No.	List of Experiments
1	Drawing of lines, horizontal lines, curved lines, lines of growth and grace, thick and thin lines, double lines, broken lines, oblique lines or lines of movement, accented lines.
2	Free hand drawing practice - creation of foliage drawing, small plants and flowers, creepers, birds, animals, clouds and still water.
3	Light and Pigment theory of colours, properties of colour vision, complementary colours, chromatic circle, attributes of colour, properties of colour – hue, value and intensity, simultaneous contrast – contrast of tone and colour harmony.
4	Modification of colours – hue, tone, tint, shade, coloured greys and triadic colour.
5	Drawing of special illusion colour contrasts – mono chromatic contrast and polychromatic contrast.
6	Free hand drawing of elements of textile design – motif, unit figure and design repeat.
7	Creation of textile designs on various bases - rectangular, diamond, ogee, diagonal bases; distribution of unit figures – principles of simple drop and drop reverse; sateen distribution, all over effects etc.
8	Creation of layout for saree, dhoti, lungi and shawl.
9	Creation of layout for bed sheet, table cover and handloom durries.
10	Documentation of traditional Indian handloom sarees. (Pictures/ samples)

BHT-652	FABRIC STRUCTURE - II LAB	0L:0T:2P	1 Credits

S. No.	List of Experiments
1	To develop the design on graph/point paper from sketch with various stages involved for production of figured single cloth.
2	To perform the punching and lacing of pattern cards from the given figured graph.
3	To create the design on graph paper and develop fabric of extra warp and extra weft on loom
4	To create the design on graph paper and develop fabric of warp backed/weft backed on loom
5	To create the design on graph paper and develop fabric of figured double cloth with 2 & 4 color effects on loom
6	To create the design on graph paper and develop fabric of figured pique on loom
7	To create the design on graph paper and develop fabric of figured terry on loom
8	To analyze the compound fabric particulars- double cloth, Bedford cloth, terry and leno
9	Analysis of weave, constructional details, weaving techniques, costing of at least two traditional handloom sarees
10	Extracting the production of particulars of given handloom jacquard design fabric and furnishing the production and cost details

BHT-653	FABRIC & GARMENT QUALITY EVALUATION LAB.	0L:0T:2P	1 Credits

S. No.	List of Experiments
1	To determine Fabric tensile strength by using strip strength method.
2	To determine the bursting strength of fabric by using Hydraulic bursting strength tester.
3	To determine the Fabric tear strength by using a (Elmendorf tear) pendulum-type tester.
4	To determine Fabric bending length, flexural rigidity and bending modulus by using stiffness tester.
5	To determine fabric Drape ability by using Cusick drape tester.
6	Measure crease recovery of fabric by crease recovery tester.
7	To determine the fabric thickness by using thickness gauge.
8	Find out the relative abrasion properties of fabrics by Martindale abrasion tester.
9	Fabric pilling resistance Test of pilling of fabrics by ICI Pilling Box Tester.
10	Find out air permeability of fabric by air permeability tester.
11	To determine the fabric GSM by using GSM cutter.